

OPERATING SYSTEMS

UE25CS242A

Segmentation

Pavan A C

Department of Computer Science & Engineering

OPERATING SYSTEMS

Slides Credits for all the PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

OPERATING SYSTEMS

Segmentation

Pavan A C
Department of Computer Science & Engineering

OPERATING SYSTEMS

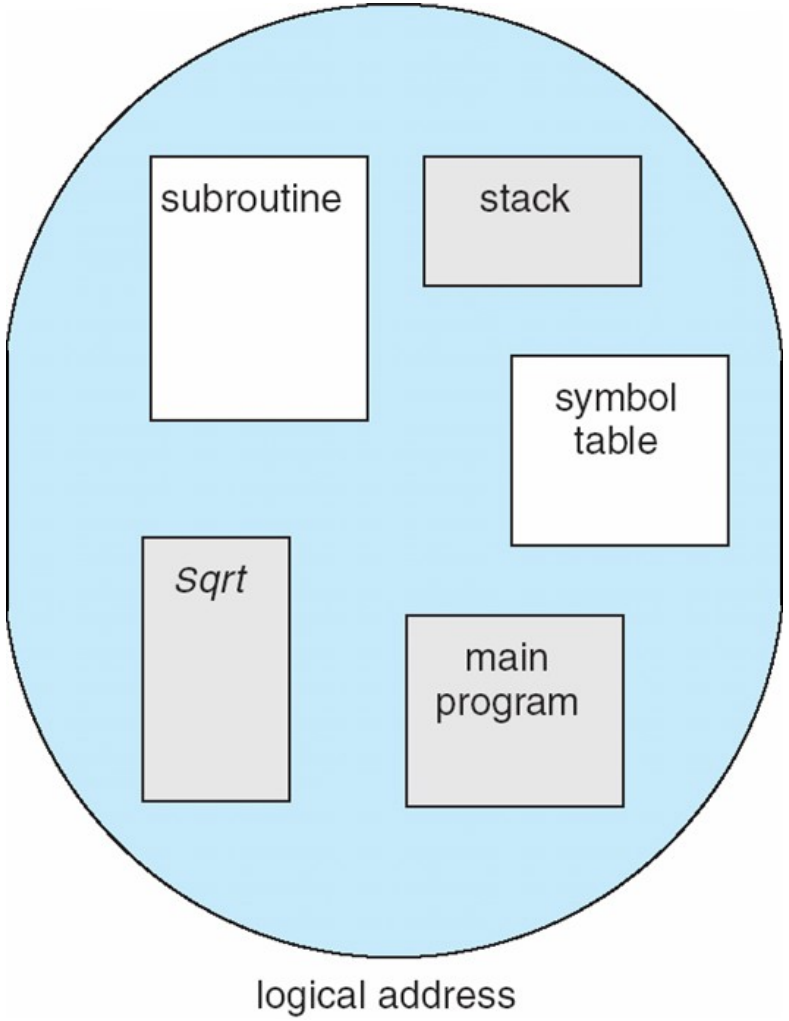
Segmentation



- Memory-management scheme that supports user view of memory
- A program is a collection of segments
  - A segment is a logical unit such as:
    main program
    procedure
    function
    method
    object
    local variables, global variables
    common block
    stack
    symbol table
    arrays

OPERATING SYSTEMS

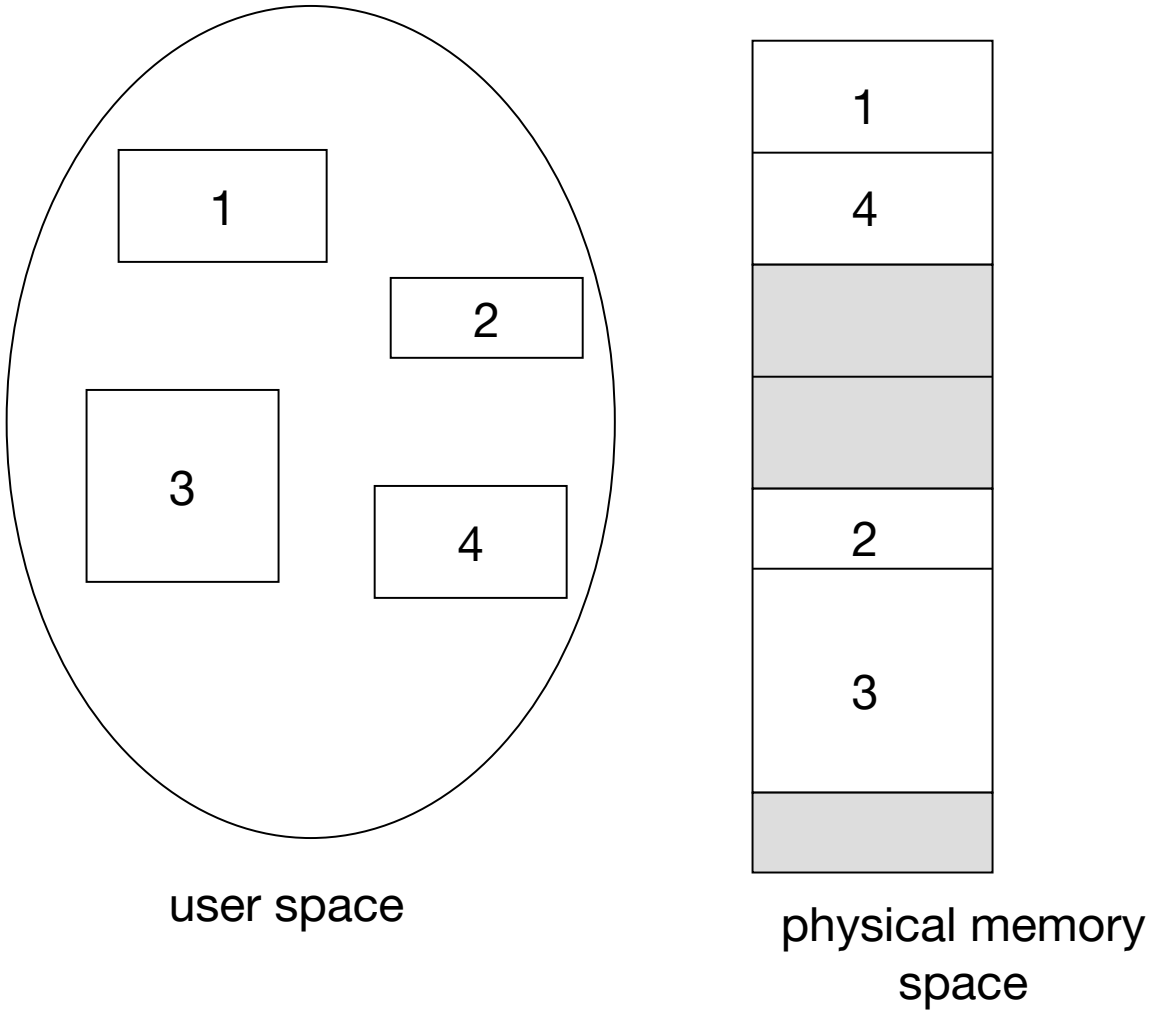
User's View of a Program



- Segmentation is a memory-management scheme that supports this programmer view of memory
- Each segment has a name and a length.
- Addresses specify segment name + offset within the segment

OPERATING SYSTEMS

Logical View of Segmentation



Segmentation Basics



- Segmentation is a technique for breaking memory up into logical pieces
- Each “piece” is a grouping of related information
  - data segments for each process
  - code segments for each process
  - data segments for the OS
  - etc.
- Segmentation permits the physical address space of a process to be non-contiguous.

Segmentation Architecture



- Logical address consists of a two tuple:

  <segment-number s, offset d>,

- **Segment table** – maps two-dimensional physical addresses; each table entry has:
  - **Segment base** – contains the starting physical address where the segments reside in memory
  - **Segment limit** – specifies the length of the segment
- **Segment-table base register (STBR)** points to the segment table's location in memory
- **Segment-table length register (STLR)** indicates number of segments used by a program;

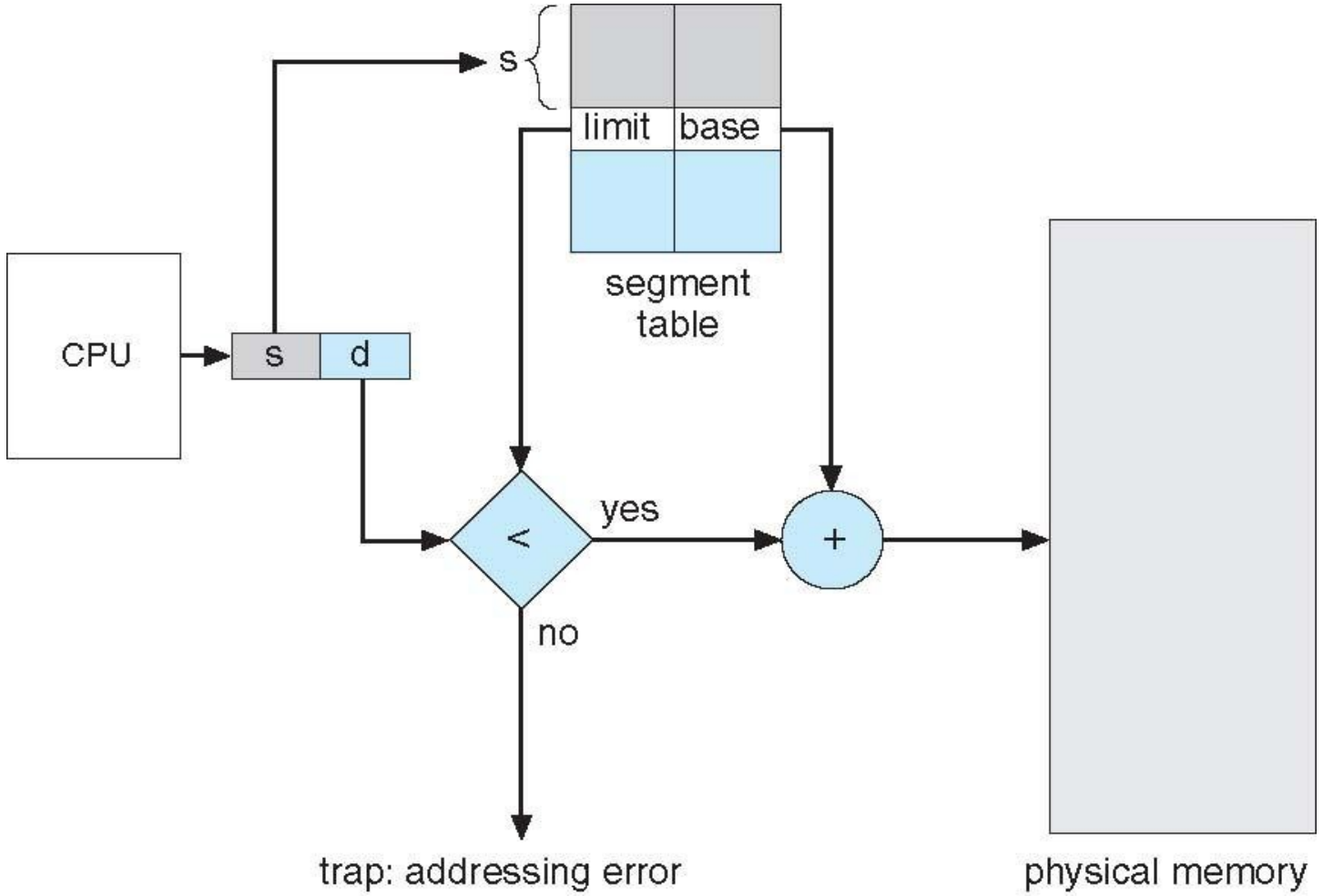
  segment number ***s*** is legal if ***s*** < **STLR**

Segmentation Hardware

- A logical address consists of two parts: a segment number, s, and an offset into that segment, d.
- The segment number is used as an index to the segment table. The offset d of the logical address must be between 0 and the segment limit. If it is not, we trap to the operating system (logical addressing attempt beyond end of segment).
- When an offset is legal, it is added to the segment base to produce the address in physical memory of the desired byte. The segment table is thus essentially an array of base–limit register pairs.

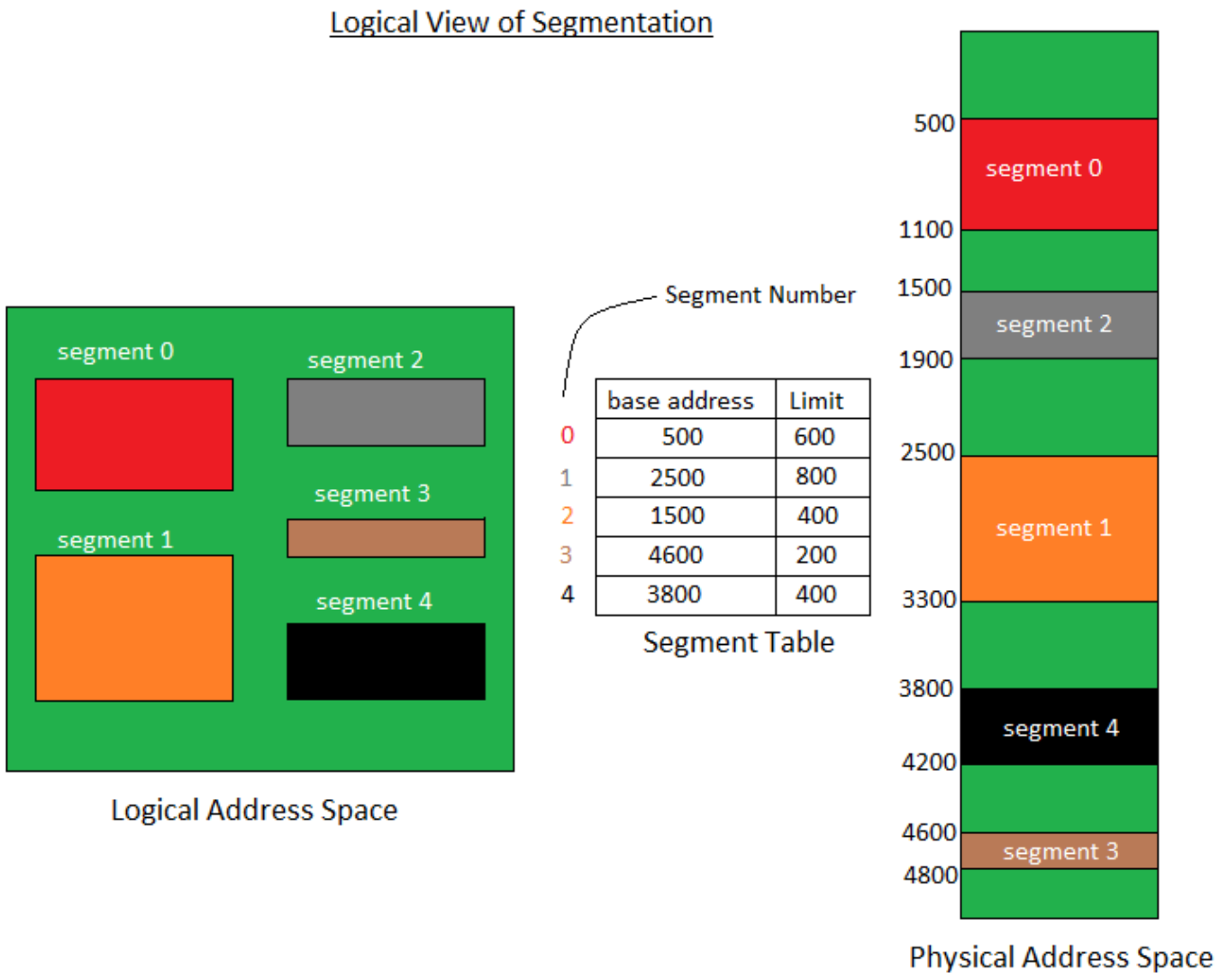
PES UNIVERSITY
CELEBRATING 50 YEARS

Segmentation Hardware (Cont.)



OPERATING SYSTEMS

Segmentation Example



Segmentation Example (Cont.)

- We have five segments numbered from 0 through 4. The segments are stored in physical memory as shown.
- The segment table has a separate entry for each segment, giving the beginning address of the segment in physical memory (or base) and the length of that segment (or limit).

PES UNIVERSITY
CELEBRATING 50 YEARS

Segmentation: Examples



- Consider the following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

a. 0,430
b. 1,10
c. 2,500
d. 3,400
e. 4,112

Segmentation: Examples

Consider the following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

a. 0,430
b. 1,10
c. 2,500
d. 3,400
e. 4,112

Solution:

a. 0, 430 →219 + 430 = 649
b. 1, 10 →2300 + 10 = 2310
c. 2, 500 Illegal address since size of segment 2 is 100 and the offset in logical address is 500.
d. 3, 400 →1327 + 400 = 1727
e. 4, 112 Illegal address since size of segment 4 is 96 and the offset in logical address is 112.

Shared Segments

- How will you implement shared memory using segmentation?
- Two processes may have segments with same base address → Segments will be mapped to both and hence shared.

PES UNIVERSITY
CELEBRATING 50 YEARS

Linux Angle

- **The Legacy of x86:** Classic Intel processors utilized explicit segmentation registers (CS for Code, DS for Data, SS for Stack).
- **Linux Flat Memory Model:** Modern Linux on 64-bit systems essentially neutralizes segmentation. The kernel sets the base address of all segments to 0x00000000 and the limit to the maximum addressable space.
- What would be most common disadvantage of Segmentation?
- External Fragmentation
- This leads us to the Paging mechanism

PES UNIVERSITY
CELEBRATING 50 YEARS

Questions



Q1: In a segmented memory system, what two components make up a logical address?
Q2: When the CPU translates a logical address in a segmented system, what causes a "trap to the operating system" (a Segmentation Fault)?
Q3: Why is it advantageous to separate the Stack and the Code into different segments?
Q4: What is the "Flat Memory Model" used by modern Linux on x86-64 architectures?
Q5: During process execution, which hardware component performs the addition of the segment base and the logical offset?



THANK YOU

Pavan A C
Department of Computer Science and Engineering

pavanac@pes.edu